🐘 HDFS Mini Project — Questions

Q1.

Create an HDFS directory named /university.

Q2.

Inside /university, create subdirectories:

- /university/cse
- /university/ece
- /university/mech

Q3.

List all directories inside /university.

Q4.

Create three local text files containing student names:

- cse.txt
- ece.txt
- mech.txt

Q5.

Upload each file into its respective HDFS folder.

Q6.

Display the contents of all uploaded files.

Q7.

Create a new local file new_students.txt and append its content to cse.txt in HDFS.

Q8.

Rename the file cse.txt to cse_2026.txt.

Q9.

Make a copy of cse_2026.txt named cse_backup.txt in the same folder.

Q10.

Move cse_backup.txt from /university/cse to /university.

Q11.

List all files inside /university and /university/cse.

Q12.

Delete the file mech.txt.

Q13.

Delete the directory /university/ece (even if it is not empty).

Q14.

Display the contents of cse_2026.txt.

Q15.

Check disk usage of /university.

Q16.

Delete the entire /university directory.

ANSWERS:

```
C:\Windows\System32>cd C:\hadoopsetup\hadoop-3.2.4\sbin

C:\hadoopsetup\hadoop-3.2.4\sbin>start-dfs.cmd

C:\hadoopsetup\hadoop-3.2.4\sbin>start-yarn.cmd
starting yarn daemons

C:\hadoopsetup\hadoop-3.2.4\sbin>jps
32032 DataNode
56512 NameNode
72484 NodeManager
51596 ResourceManager
60508 Jps

C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -ls /
Found 2 items
drwxr-xr-x   - supradeep supergroup          0 2026-03-16 14:47 /data
drwxr-xr-x   - supradeep supergroup          0 2026-03-16 14:45 /test

C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -mkdir /university

C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -mkdir /university/cse

C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -mkdir /university/ece

C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -mkdir /university/mech
```

```
C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -ls /university
Found 3 items
drwxr-xr-x   - supradeep supergroup          0 2026-03-16 15:39 /university/cse
drwxr-xr-x   - supradeep supergroup          0 2026-03-16 15:39 /university/ece
drwxr-xr-x   - supradeep supergroup          0 2026-03-16 15:40 /university/mech

C:\hadoopsetup\hadoop-3.2.4\sbin>notepad cse.txt

C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -put cse.txt /university/cse

C:\hadoopsetup\hadoop-3.2.4\sbin>notepad ece.txt

C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -put ece.txt /university/ece

C:\hadoopsetup\hadoop-3.2.4\sbin>notepad mech.txt

C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -put mech.txt /university/mech

C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -cat /university/cse/cse.txt
1
2
3

C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -cat /university/ece/ece.txt
4
5
6

C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -cat /university/mech/mech.txt
7
8
```

9

```
C:\hadoopsetup\hadoop-3.2.4\sbin>notepad new_students.txt

C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -appendToFile new_students.txt
/university/cse/cse.txt

C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -mv /university/cse/cse.txt
/university/cse/cse_2026.txt

C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -cp /university/cse/cse_2026.txt
/university/cse/cse_backup.txt

C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -mv /university/cse/cse_backup.txt /university/

C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -ls /university
Found 4 items
drwxr-xr-x   - supradeep supergroup          0 2026-03-16 15:44 /university/cse
-rw-r--r--   1 supradeep supergroup          9 2026-03-16 15:44 /university/cse_backup.txt
drwxr-xr-x   - supradeep supergroup          0 2026-03-16 15:41 /university/ece
drwxr-xr-x   - supradeep supergroup          0 2026-03-16 15:42 /university/mech

C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -ls /university/cse
Found 1 items
-rw-r--r--   1 supradeep supergroup          9 2026-03-16 15:43 /university/cse/cse_2026.txt

C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -rm /university/mech/mech.txt
Deleted /university/mech/mech.txt

C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -rm -r /university/ece
Deleted /university/ece

C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -cat /university/cse/cse_2026.txt
```

1

2

3

```
C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -du /university
9  9  /university/cse
9  9  /university/cse_backup.txt
0  0  /university/mech

C:\hadoopsetup\hadoop-3.2.4\sbin>hdfs dfs -rm -r /university
Deleted /university

C:\hadoopsetup\hadoop-3.2.4\sbin>
```